
O-RAN Work Group 6

Application Service Descriptor specification

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O-RAN ALLIANCE e.V., Buschkauler Weg 27, 53347 Alfter, Germany

Register of Associations, Bonn VR 11238, VAT ID DE321720189

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Foreword

This Technical Specification (TS) has been produced by WG6 of the O-RAN Alliance.

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN Alliance modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

version xx.yy.zz

where:

- xx: the first digit-group is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc. (the initial approved document will have xx=01). Always 2 digits with leading zero if needed.
- yy: the second digit-group is incremented when editorial only changes have been incorporated in the document. Always 2 digits with leading zero if needed.
- zz: the third digit-group included only in working versions of the document indicating incremental changes during the editing process. External versions never include the third digit-group. Always 2 digits with leading zero if needed.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

1 Scope

The present document specifies the structure and format of the Application Service Descriptor (ASD).

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

- [1] Kubernetes label key-value-nomenclature <https://kubernetes.io/docs/concepts/overview/working-with-objects/labels/>
- [2] O-RAN-WG6.ORCH-USE-CASES-R003: "Cloudification and Orchestration Use Cases and Requirements for O-RAN Virtualized RAN".
- [3] IETF RFC 4122: "A Universally Unique IDentifier (UUID) URN Namespace".
- [4] OASIS "TOSCA Simple Profile in YAML Version 1.3"

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

- [i.1] O-RAN.WG6.AppLCM-Deployment-R003-v01.00: "Application Life Cycle Management (LCM) for Deployment Technical Recommendation".
- [i.2] O-RAN.WG10.OAM-Architecture-R003: "O-RAN Operations and Maintenance Architecture".
- [i.3] Kubernetes node features <https://kubernetes-sigs.github.io/node-feature-discovery/stable/get-started/index.html>

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in [2] and the following apply:

Descriptor: set of attributes with values that contains information necessary to lifecycle manage an application or a resource, expressed in a template.

NOTE: The term Descriptor is always qualified, explicitly or implicitly by the context, to refer to a specific type, like Application Service Descriptor, External Connection Point Descriptor, among others.

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

ASD	Application Service Descriptor
CNI	Container Network Interface
DC	Data Center
DPDK	Data Plane Development Kit
ExtCpd	External Connection Point Descriptor
GARP	Gratuitous ARP (Address Resolution Protocol)
GPU	Graphics Processing Unit
IPAM	IP Address Management
K8s	Kubernetes
LAG	Link Aggregation Group
LI	Lawful Interception
MAC	Media Access Control
NAD	Network Attachment Definition
NF	Network Function
NW	Network
OAM	Operations, Administration and Management
O-RAN	Open Radio Access Network
OVS	Open vSwitch
PCIe	Peripheral Component Interconnect Express
PF	Physical Function
SMO	Service Management and Orchestration
SR-IOV	Single Root I/O Virtualization
URI	Uniform Resource Identifier
UUID	Universally Unique Identifier
VLAN	Virtual Local Area Network
VF	Virtual Function
vNIC	virtual Network Interface Card

VPN Virtual Private Network
VPP Vector Packet Processing

4 Objective

The present document delivers a specification of the information elements and attributes applicable to the ASD and a specification of a TOSCA data model of the ASD.

5 Conventions

The attributes of the ASD information elements are described in the tables provided in section 6.2. Each table has 5 columns, with the following significance:

- The "Attribute" column provides the attribute name.
- The "Qualifier" column indicates whether the support of the attribute is mandatory, optional or conditional from the SMO perspective.
- The "Cardinality" column contains the minimum and maximum cardinality of this information element (e.g., 1, 0 ... N, 1 ... N). A cardinality range starting with 0 indicates that the attribute need not always be included and 1 indicates that the attribute needs to be always included.
- The "Type" column provides information on the type of the attribute values. It can be the name of an Information Element. If a cell in the "Type" is marked as "Not specified", which indicates that the specification of the type is left to the data model.
- The "Description" provides a brief explanatory description, additional constraints, and examples.

6 Application Service Descriptor information elements

6.1 Overview

The Application Service Descriptor (ASD) contains data that describes an application and that is used by the SMO as input for the deployment of the application workload.

The ASD is a complementary descriptor to the cloud-native deployment artifacts. Hence, it contains design-time information provided by the application vendor that is not described/available in the cloud native artifacts. Such complementary information in the ASD can include specific infrastructure capabilities requirements for the deployment of the application software workload, constraints on the Kernel type needed, additional networking requirements needed beyond primary connectivity described in the cloud native artifacts, any special CNI (Container Network Interfaces), etc [i.1].

The ASD can support an application workload deployment on OS container infrastructures. It is only applicable to O-RAN NF Deployments.

Figure 6.1-1 illustrates the Application Service Descriptor and its deployment related artifacts: the CloudNativeDescriptorFile and the ExecutableImageFile.

Figure 6.1-2 shows how the ASD is positioned within the NF Deployment Package Model.

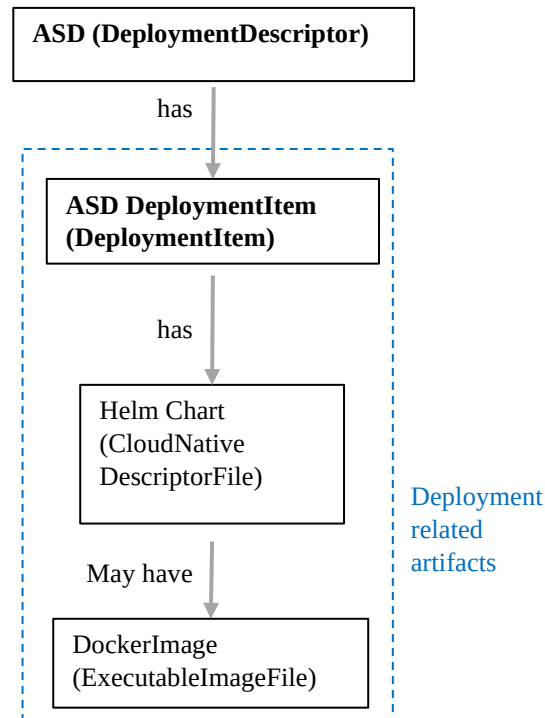


Figure 6.1-1: DeploymentDescriptor with ASD representation

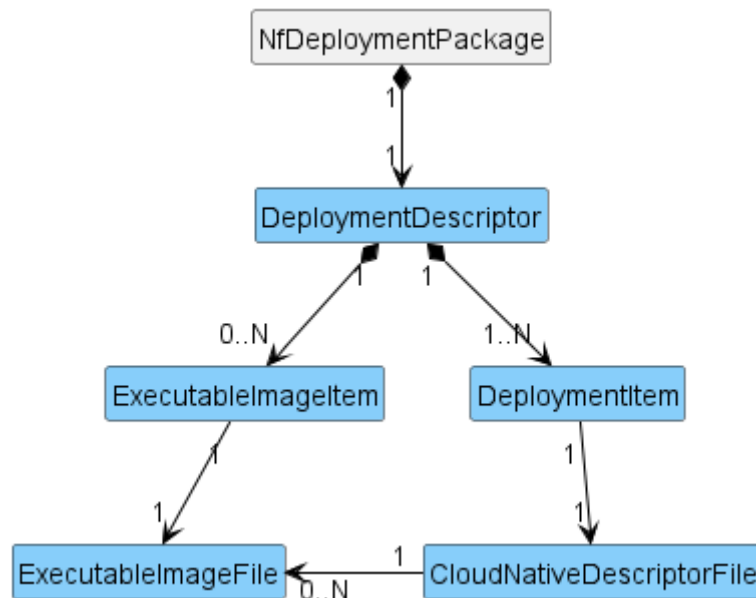


Figure 6.1-2: ASD descriptor and deployment artifacts within the NF Deployment Package

6.2 Application Service Descriptor Information Element

For the O-RAN NF Deployments the Application Service Descriptor (ASD) is a main top level descriptor template which consists of the attributes described in Table 6.2-1 below.

Table 6.2-2: ASD Information Model

Attribute	Qualifier	Cardinality	Type	Description
asdId	M	1	String	Identifier of this ASD information element. This attribute shall be globally unique. The format will be defined in the data model specification phase.
asdVersion	M	1	String	Identifies the version of the ASD. See note 1.
asdSchemaVersion	M	1	String	Specifies the version of the ASD's schema, i.e. the version of this specification the ASD is compliant with.
asdProvider	M	1	String	Provider of the application and of the ASD.
asdApplicationName	M	1	String	Name to identify the application. Invariant for the application lifetime.
asdApplicationSwVersion	M	1	String	Specifies the version of the software of the application. See note 2.
asdApplicationInfoName	M	0..1	String	Human readable name for the application. Can change during the ASD lifetime.
asdInfoDescription	M	0..1	String	Human readable description of the application. Can change during the ASD lifetime.
asdExtCpd	M	0..N	ExtCpd	Describes the externally exposed connection points of the application.
enhancedClusterCapabilities	M	0..1	EnhancedClusterCapabilities	A list of expected capabilities of the target deployment cluster to aid placement of the application on a suitable cluster.
deploymentItem	M	1..N	DeploymentItem	Cloud native deployment artifacts associated with the application.
executableImageItem	M	0..N	ExecutableImageItem	Contains data (e.g. name, location) related to an executable software image used to implement the application.
asdInvariantId	M	1	String	Identifier of this descriptor in a version independent manner. This attribute is invariant across versions of ASD. It is in UUID format as specified in IETF RFC 4122 [3].
NOTE 1: Version changes when a change is made to any one of the ASD attributes.				
NOTE 2: Version changes when software (software images or deployment items artifacts) of the ASD changes.				

The asdVersion changes whenever anything in the ASD changes. That includes changes in the software images or in the deployment artifacts, as changes in those items are reflected in changes in ASD attributes.

The `asdApplicationSwVersion` changes only when there are changes in the software of the application described by the ASD, i.e. changes in the software images or in the deployment artifacts. Thus, a change in the `asdApplicationSwVersion` implies a change in the `asdVersion`, but not vice versa.

The `asdSchemaVersion` indicates the version of the standard the ASD is compliant with, e.g. 2.0.0. It allows the entity processing the ASD to correctly parse and interpret its values. It can be noted that a change in the `asdSchemaVersion`, e.g. to indicate compliance with a newer version of the standard, implies a change in the `asdVersion`, since it is a change in an ASD attribute.

Changes in any of the above three attributes do not result in a change in the `asdInvariantId` as long as the ASD is describing the same application. It is expected that the `asdInvariantId` rarely changes. It is up to the application vendor to define how it changes.

6.3 Deployment Item (DeploymentItem) Information Element

The `deploymentItem` defined in O-RAN Operations and Maintenance Architecture [i.2] is realized by the `DeploymentItem` IE which has the `DeploymentItem` structure described in Table 6.3-1 below.

The `DeploymentItem` IE describes the type of cloud native artifact, its order of deployment in case multiple cloud native artifacts are employed with the same ASD, and the list of parameters that can be parameterized in this cloud native deployment artifact at deployment time.

Table 6.3-1: DeploymentItem IE

Attribute	Qualifier	Cardinality	Type	Description
<code>deploymentItemId</code>	M	1	String	The identifier of this deployment item
<code>artifactType</code>	M	1	String	Specifies the artifact type. One of following values can be chosen: "helm_chart", "helmfile", "crd", "terraform".
<code>artifactId</code>	M	1	String	Unique reference to a deployment item (cloud native deployment artifact). It can be a URI or file path.
<code>deploymentOrder</code>	M	0..1	Integer	Specifies the deployment order of the deployment item. A lower value specifies that the deployment item belongs to an earlier deployment stage, i.e., needs to be installed prior to a deployment item with higher <code>deploymentOrder</code> values. If not specified, the deployment of the deployment item can be done in arbitrary order and decided by the SMO.
<code>deploymentParameter</code>	M	0..N	String	The list of parameters that can be overridden at deployment time (e.g., the list of parameters in the <code>values.yaml</code> which can be overridden at deployment time)

6.4 External connection point descriptor (ExtCpd) Information Element

The `ExtCpd` IE describes endpoints exposed by the application to enable orchestrators to string together or optimally place linked applications.

Table 6.4-2: ExtCpd IE

Attribute	Qualifier	Cardinality	Content	Description
id	M	1	String	The identifier of this ExtCpd.
description	M	1	String	Describes the service exposed.
virtualLinkRequirement	M	1	String	Refers in an abstract way to the network that the external connection point shall be exposed for (for example, OAM, EndUser, backhaul, LI, etc). The intent is to enable a network operator to identify which external connection point was designed to be connected to which VPN.
networkInterfaceRealizationRequirements	M	0..1	NetworkInterfaceRealizationRequirements	Details container implementation specific requirements on the NetworkAttachmentDefinition. See note 1.
inputParamMappings	M	0..1	ExtCpdParamMappings	Information on what parameters are required to be provided to the deployment tools for the external connection point.
resourceMapping	M	0..1	String	Resource name for the cloud native resource manifest, as specified in the cloud native descriptor declaring the network interface (e.g., in the helm chart, for the: service, ingress or pod resource). Enables, together with knowledge on namespace, the SMO to lookup the runtime data related to the external connection point. See note 2.
NOTE 1: Applies only for ExtCpd representing secondary network interfaces in a POD.				
NOTE 2: The format of the string is specific for each different orchestration templating technology used (Helm, Terraform, etc.). Currently only the format for use with Helm charts is specified: helmchartname:resourceName.				

6.5 NetworkInterfaceRealizationRequirements Information Element

The NetworkInterfaceRealizationRequirements IE describes details related to secondary networks that attach the OS containers to the logical or physical networks.

Table 6.5-1 NetworkInterfaceRealizationRequirements IE

Attribute	Qualifier	Cardinality	Content	Description
trunkMode	M	0..1	Boolean	Specifies whether the interface is capable of carrying traffic for multiple VLANs. If not present or set to false: this interface shall connect to single network. If set to true: the network interface shall be a trunk interface (connects to multiple VLANs).
ipam	M	0..1	Enum	Specifies which mode is used for the IP address assignment.

Attribute	Qualifier	Cardinality	Content	Description
				Values: "infraProvided": the CNI specifies how IPAM is done and assigns the IP address to the pod interface. "orchestrated": the IP address to be assigned to the pod interface is expected from the orchestration function as input parameter to the deployment request. "userManaged": the IP address is provided directly to the application via an external management entity. Default value is "infraProvided".
interfaceType	M	0..1	Enum	Specifies the type of network interface. Different types of interfaces are supported by different CNIs. Values: "kernel.netdev": Linux kernel or vSwitch based. It is supported by CNIs™ like: OVS, MACVLAN, IPVLAN, BRIDGE, PTP, etc. "direct.userspace": Provides direct access to network PCIe, typically an SR-IOV VF, but could also be a PF. Uses application specific driver contained in the image. The device is bound to a driver by the cloud provider which supports user space control by the application (e.g. vfio_pci, igb_uio). It is supported by the following CNIs™: SRIOV, HOST-DEVICE. "direct.kerneldriver": Similar to direct.userspace but device is bound to a kernel driver by the cloud provider and can be consumed in the same way as a kernel netdev. It is supported by the following CNIs™: SRIOV, HOST-DEVICE. "direct.bond": Based on the BOND CNI™. Requires the existence of a mated pair of direct kernel network attachments prior to the creation of the bond. Thus, when this interface type is indicated, three attachments definitions to the secondary container cluster network are expected: two providing the direct kerneldriver functionality and one with the BOND CNI™

Attribute	Qualifier	Cardinality	Content	Description
				"userspace": Based on the userspace CNI™ requires infrastructure to support a DPDK-OVS or VPP based vSwitch. Default value is "kernel.netdev".
interfaceOption	M	0..N	Enum	vNIC configurations the network interface is verified to work with. This attribute is applicable for interfaces of type "userspace". Values: "virtio" "memif"
interfaceRedundancy	M	0..1	Enum	Method required from the infrastructure to provide redundancy for the interface. Values: "infraProvided", "left", "right", "activeActiveBond", "activePassiveBond", "activePassive3" "infraProvided": The application sees one vNIC, but the infrastructure provides redundant access to the network via both switch planes. For interface of type kernel.netdev the redundancy is provided by the vSwitch or Linux bonding. For interfaces of type direct, it requires that the physical NIC is connected to both switch planes (a.k.a. smartNIC). Thus, it imposes hardware requirements on the infrastructure. For interface of type userspace redundancy handled by a DPDK-OVS or VPP vSwitch "left" and "right": indicates a vNIC connected non-redundantly to the network via one specific (left or right) switchplane. Redundancy is implemented by having two external connection points, indicating different values (left/right) in the descriptor. For the the following values, the infrastructure provides for a single external connection point a mated vNIC pair in the Pod, one connecting to the network via left switchplane and the other connecting to the network via the right switchplane, and with the application using them together as a redundant network interface using a particular redundancy method that

Attribute	Qualifier	Cardinality	Content	Description
				need to be accomodated in the node infrastructure: "activeActiveBond": Requires a static multi-chassis LAG in active-active mode from the infrastructure. Typically corresponds to bonding mode-2 "balance-xor" "activePassiveBond": The interfaces are bonded in active-passive mode in the application with move of bond MAC address. No specific requirements on DC fabric. Typically corresponds to bonding mode-1 "active-backup" "activePassive13": Application moves its IP address by attaching it to one of the two MACs by use of GARP messages.

6.6 ExtCpdParamMappings Information Element

The ExtCpdParamMappings IE describes required information on what parameters to be provided to the deployment tools for the ExtCpd.

Table 6.6-1: ExtCpdParamMappings IE

Attribute	Qualifier	Cardinality	Content	Description
loadbalancerIP	M	0..1	String	When present, this attribute specifies the name of the deployment artifact input parameter through which the SMO can configure the loadbalancerIP parameter of the K8s service or ingress controller that the ExtCpd represents. The param name and provided IP address value will be passed to the deployment tool when deploying the deployment artifacts. See note 2.
externalIPs	M	0..1	String	When present, this attribute specifies the name of the deployment artifact input parameter through which the SMO can configure the externalIPs parameter of the K8s service or ingress controller, or the pod network interface annotation, that the ExtCpd represents. The param name and provided IP address(es) value will be passed to the deployment tool when deploying the deployment artifacts. See note 2.
nadName	M	0..N	String	These attributes specify, for an ExtCpd representing a secondary network interface, the name(s) of the deployment artifact input parameter(s) through which the SMO can provide the names of the network attachment definitions (NADs) the SMO has created as base for the network interface the ExtCpd represents. The param name(s) and provided NAD name value(s) will be passed to the deployment tool when deploying the deployment artifacts. It is expected that the NADs themselves have been created prior to the deployment of the deployment artifacts. See notes 1, 2, and 3.
nadNamespace	M	0..1	String	Specifies, for an ExtCpd representing a secondary network interface, the name of the deployment artifact input parameter through which the SMO can provide the namespace where the NADs are located. The param name and provided namespace value will be passed to the deployment tool when

Attribute	Qualifier	Cardinality	Content	Description
				deploying the deployment artifacts. Attribute may be omitted if the namespace is the same as the application namespace. See note 2.
NOTE 1: When the ExtCpd represent a network redundant/mated pair of sriov interfaces, there are references to 2 or 3 related NADs needed to be passed, while for other interface types only one NAD reference is needed to be passed. NOTE 2: The format of the string content is specific for each different orchestration templating technology used (Helm, Teraform, etc.). Currently only a format for use with Helm charts is specified: "<helmchartname>:[<subchartname>.] 0 ... N[<parentparamname>.] 0 ... N<paramname>". Whether the optional parts of the format are present depends on how the parameter is declared in the helm chart. An example is: "chartName:subChart1.subChart2.subChart3.Parent1.Parent2.Parent3.parameter". NOTE 3: A direct attached (passthrough) network interface, such as a sriov interface, attaches to a network via only one of the two switch planes in the infrastructure. When using a direct attached network interface one therefore commonly in a pod uses a mated pair of sriov network attachments, where each interface attaches to the same network but via a different switchplane. The application uses the mated pair of network interfaces as a single logical "switch-path-redundant" network interface – and this is represented by a single ExtCpd. Also, there is a case where a third "bond" attachment interface is used in the pod, bonding the two direct interfaces so that the application does not need to handle the redundancy issues – application just uses the bond interface. In this case all three attachments are together making up a logical "switch-path-redundant" network interface represented by a single ExtCpd. When three NADs are used in the ExtCpd the NAD implementing the bond attachment interface is provided through the parameter indicated in the third place in the nadName attribute.				

6.7 EnhancedClusterCapabilities Information Element

The EnhancedClusterCapabilities IE describes information which is used to aid placement of the application service on a suitable cluster.

Table 6.7-1: EnhancedClusterCapabilities IE

Attribute	Qualifier	Cardinality	Content	Description
minKernelVersion	M	1	String	Describes the minimal required Kernel version, e.g., 4.15.0. Coded as displayed by linux command uname -r
requiredKernelModule	M	0..N	String	Required kernel modules are coded as listed by linux lsmod command, e.g., ip6_tables, cryptd, nf_nat etc.
conflictingKernelModule	M	0..N	String	Kernel modules, which shall not be present in the target environment. The kernel modules are coded as listed by linux lsmod command, e.g., ip6_tables, cryptd, nf_nat etc. Example: Linux kernel SCTP module, which may conflict with use of proprietary user space SCTP stack provided by the application.
requiredCustomResource	M	0..N	Structure (inlined)	List of the required custom resource types in the target environment, identifying each by the "kind" and "apiVersion" field in the K8S resource manifests and in the application. The list shall include those custom resource types which are not delivered with the application. Example: requiredCustomResources: - {kind: "Redis", apiVersion: " kubedb.com/v1alpha1 "}
>kind	M	1	String	Kind of the custom resource

Attribute	Qualifier	Cardinality	Content	Description
>apiVersion	M	0..1	String	apiVersion of the custom resource. If not indicated, any apiVersion of the custom resource is valid.
clusterLabel	M	0..N	String	This attribute indicates the required O-Cloud Node Cluster capabilities. These can indicate special infrastructure capabilities (e.g., NW acceleration, GPU compute, etc.). The intent of these O-Cloud Node Cluster capabilities is to serve as a set of values that can support the SMO in application placement decisions. See note 1. clusterLabel follow the Kubernetes label key-value-nomenclature [1] . This attribute should use the standardized values for the O-RAN O-Cloud Node Cluster capabilities, the existing values for node features [i.3], and any new capabilities identified in O-RAN specifications. The attribute supports also extensions for non standardized values. See note 2.
requiredPlugin	M	0..N	Structure (inlined)	A list of the names and versions of the required K8s plugin (e.g., multus v3.8)
>requiredPluginName	M	1	String	The names of the required K8s plugin (e.g., multus)
>requiredPluginVersion	M	0..1	String	The version of the required plugin (e.g., 3.8). If not indicated, any version of the plugin is valid.
NOTE 1: The SMO uses the set of standardized capabilities to identify O-Cloud Node cluster capabilities required in the DeploymentDescriptors and to match them to the capabilities available in the O-Cloud. NOTE 2: References to standandardized values for the O-RAN O-Cloud Node Cluster capabilities and/or new capabilities identified in O-RAN specifications will be added in future versions of the present document.				

6.8 Executable Image Item (ExecutableImageItem) Information Element

The ExecutableImageItem IE contains metadata related to an executable software image and the reference to its location.

Table 6.8-3: ExecutableSoftwareItem IE

Attribute	Qualifier	Cardinality	Type	Description
swImageName	M	1	String	Name of the software image
swImageVersion	M	1	String	Version of the software image
swImageLocation	M	1	String	URI (a file path, external URL, etc.) referring to the location where the image file is located.

7 TOSCA data model of the ASD

7.1 General concept of using TOSCA to model the ASD

The ASD shall be written as a TOSCA YAML service template using the grammar specified in TOSCA Simple Profile in YAML v1.3 [4].

However, only a limited set of the TOSCA modelling entities and grammar features specified in [4] are supported in an O-RAN compliant ASD. Clause 7.2 lists entities and features excluded from the ASD.

Clauses 7.3, 7.4 and 7.5 specifies O-RAN data types, artifact types and node types to be used in an ASD. These clauses also list the keynames that shall, may or shall not be included in the vendor specific node template and artifact definitions. Definition of additional, vendor specific, types, is not supported.

Annex A contains the formal, machine readable, YAML definitions of the data types, artifact types and node types specified in clauses 7.3, 7.4 and 7.5. The Annex A is considered the source of truth. Thus, in case of any discrepancy between the definitions in clauses 7.3-7.5 and the YAML definitions in the annex A, the latter ones take precedence.

Clause 7.6 specifies the structure of the TOSCA ASD service template, detailing which elements it consists of and whether they are mandatory or optional.

7.2 Restrictions in the use of TOSCA for the ASD model

The file with the TOSCA specification of the ASD is called a service template. The service template is the DeploymentDescriptor in figure 6.1-2. It contains a “topology template” as its main component. As described in [4] the topology template is a TOSCA construct that represents a graph of the node templates modeling the components a workload is made up of and the relationships between them.

The ASD topology template consists of one single node template that contains one or several artifacts. The artifacts are the DeploymentItem and the ExecutableImageItem in figure 6.1-2. The structure of the topology template and the complete service template is described in clause 7.6. All other entities and features available in the TOSCA language are not supported in the ASD. In particular, the following features are excluded:

- Requirements, capabilities and relationships
- Groups
- Policies (events, triggers, conditions, actions, etc)
- Interfaces (operations, notifications, scripts, etc.)
- Workflows
- Template substitution (substitution mappings)
- Functions
- Inputs/Outputs
- dsl_definitions (definition of reusable macros for use throughout the service template)
- Attributes
- Node filter

The above list is not intended to be exhaustive. Clause 7.6 determines what TOSCA elements are used.

7.3 Data types

7.3.1 ExtCpdData

7.3.1.1 Description

Data type name: **tosca.datatypes.asd.ExtCpdData**

This data type describes the externally exposed connection points of the application, as defined in clause 6.4.

7.3.1.2 Properties

The properties of the **tosca.datatypes.asd.ExtCpdData** data type are specified in table in table 7.3.1.2-1.

Table 7.3.1.2-1: Properties

Name	Required	Type	Constraints	Description
id	yes	string		The identifier of the ExtCpd
description	yes	string		Describes the service exposed by the external connection point represented by this ExtCpd.
virtualLinkRequirement	yes	string		Refers in an abstract way to the network that the external connection point shall be exposed for (for example, OAM, EndUser, backhaul, LI, etc). The intent is to enable a network operator to identify which external connection point was designed to be connected to which VPN.
networkInterfaceRealizationRequirements	no	tosca.datatypes.asd.NetworkInterfaceRequirements		Details container implementation specific requirements on the NetworkAttachmentDefinition. See note 1.
inputParamMappings	no	tosca.datatypes.asd.ExtCpdParamMappings		Information on what parameters are required to be provided to the deployment tools for the external connection point.
resourceMapping	no	string		Resource name for the cloud native resource manifest, as specified in the cloud native descriptor declaring the network interface (e.g., in the helm chart, for the: service, ingress or pod resource). Enables, together with knowledge on namespace, the SMO to lookup the runtime data related to the external connection point. See note 2.

NOTE 1: Applies only for external connection points representing secondary network interfaces in a POD.

NOTE 2: The format of the string is specific for each different orchestration templating technology used (Helm, Terraform, etc.). Currently only the format for use with Helm charts is specified: helmchartname:resourcename.

7.3.1.3 Definition

The syntax of the **tosca.datatypes.asd.ExtCpdData** is defined in annex A.

7.3.2 EnhancedClusterCapabilities

7.3.2.1 Description

Type name: **tosca.datatypes.asd.EnhancedClusterCapabilities**

This data type describes expected capabilities of the target deployment cluster, as defined in clause 6.7.

7.3.2.2 Properties

The properties of the `tosca.datatypes.asd.EnhancedClusterCapabilities` data type are specified in table 7.3.2.2-1.

Table 7.3.2.2-1: Properties

Name	Required	Type	Constraints	Description
<code>minKernelVersion</code>	yes	string		Describes the minimal required Kernel version, e.g., 4.15.0. Coded as displayed by linux command <code>uname -r</code> .
<code>requiredKernelModule</code>	no	list of string		Required kernel modules are coded as listed by linux <code>lsmod</code> command, e.g., <code>ip6_tables</code> , <code>cryptd</code> , <code>nf_nat</code> etc.
<code>conflictingKernelModule</code>	no	list of string		Kernel modules, which shall not be present in the target environment. The kernel modules are coded as listed by linux <code>lsmod</code> command, e.g., <code>ip6_tables</code> , <code>cryptd</code> , <code>nf_nat</code> etc. Example: Linux kernel SCTP module, which may conflict with use of proprietary user space SCTP stack provided by the application.
<code>requiredCustomResource</code>	no	list of <code>tosca.datatypes.asd.RequiredCustomResource</code>		List of the required custom resource types in the target environment, identifying each by the "kind" and "apiVersion" field in the K8S resource manifests and in the application. The list shall include those custom resource types which are not delivered with the application. Example: kind: "Redis" apiVersion: "kubedb.com/v1alpha1"
<code>clusterLabel</code>	no	list of string		Indicates the required O-Cloud Node Cluster capabilities. These can indicate special infrastructure capabilities (e.g., NW acceleration, GPU compute, etc.). The intent of these O-Cloud Node Cluster capabilities is to serve as a set of values that can support the SMO in application placement decisions. See note 1. <code>clusterLabel</code> follow the Kubernetes label key-value-nomenclature [1] . This attribute should use the standardized values for the O-RAN O-Cloud Node Cluster capabilities, the existing values for node features [i.3], and any new capabilities identified in O-RAN specifications. The attribute supports also extensions for non standardized values. See note 2.
<code>requiredPlugin</code>	no	list of <code>tosca.datatypes.asd.RequiredPlugin</code>		A list of the names and versions of the required K8s plugin (e.g., <code>multus v3.8</code>)

NOTE 1: The SMO uses the set of standardized capabilities to identify O-Cloud Node cluster capabilities required in the `DeploymentDescriptors` and to match them to the capabilities available in the O-Cloud.

NOTE 2: References to standandardized values for the O-RAN O-Cloud Node Cluster capabilities and/or new capabilities identified in O-RAN specifications will be added in future versions of the present document.

7.3.2.3 Definition

The syntax of the `tosca.datatypes.asd.EnhancedClusterCapabilities` data type is defined in annex A.

7.3.3 NetworkInterfaceRequirements

7.3.3.1 Description

Data type name: **`tosca.datatypes.asd.NetworkInterfaceRequirements`**

This data type describes details related to secondary networks that attach the OS containers to the logical or physical networks, as defined in clause 6.5.

7.3.3.2 Properties

The properties of the `tosca.datatypes.asd.NetworkInterfaceRequirements` data type are specified in table in table 7.3.3.2-1.

Table 7.3.3.2-1: Properties

Name	Required	Type	Constraints	Description
<code>trunkMode</code>	yes (see note)	boolean	default: false	Specifies whether the interface is capable of carrying traffic for multiple VLANs. If not present or set to false: this interface shall connect to single network. If set to true: the network interface shall be a trunk interface (connects to multiple VLANs).
<code>ipam</code>	yes (see note)	string	Valid values: See YAML definition constraints default: <code>infraProvided</code>	Specifies which mode is used for the IP address assignment, as specified in clause 6.5.
<code>interfaceType</code>	yes (see note)	string	Valid values: See YAML definition constraints default: <code>kernel.netdev</code>	Specifies the type of network interface, as specified in clause 6.5. Different types of interfaces are supported by different CNIs.
<code>interfaceOption</code>	no	list of string	Valid values: See YAML definition constraints	vNIC configurations the network interface is verified to work with. This attribute is applicable for interfaces of type <code>userspace</code> .
<code>interfaceRedundancy</code>	no	string	Valid values: See YAML definition constraints	Method required from the infrastructure to provide redundancy for the interface, as specified in clause 6.5.

NOTE: This property shall exist and have a defined value whenever the `tosca.datatypes.asd.NetworkInterfaceRequirements` is used, e.g. in a node template. However, since it has a default value, an explicit property assignment is not mandatory, e.g. in a node template. If no explicit property assignment is present, the default value is assigned.

7.3.3.3 Definition

The syntax of the `tosca.datatypes.asd.NetworkInterfaceRequirements` is defined in annex A.

7.3.4 ExtCpdParamMappings

7.3.4.1 Description

Type name: **`tosca.datatypes.asd.ExtCpdParamMappings`**

This data type describes required information on what parameters to be provided to the deployment tools for the ExtCpd instance, as defined in clause 6.6.

7.3.4.2 Properties

The properties of the `tosca.datatypes.asd.ExtCpdParamMappings` data type are specified in table 7.3.4.2-1.

Table 7.3.4.2-1: Properties

Name	Required	Type	Constraints	Description
loadbalancerIP	no	string		When present, this attribute specifies the name of the deployment artifact input parameter through which the SMO can configure the loadbalancerIP parameter of the K8s service or ingress controller that the ExtCpd represents. The param name and provided IP address value will be passed to the deployment tool when deploying the deployment artifacts. See note 2.
externalIPs	no	string		When present, this attribute specifies the name of the deployment artifact input parameter through which the SMO can configure the externalIPs parameter of the K8s service or ingress controller, or the pod network interface annotation, that the ExtCpd represents. The param name and provided IP address(es) value will be passed to the deployment tool when deploying the deployment artifacts. See note 2.
nadName	no	list of string		These attributes specify, for an ExtCpd representing a secondary network interface, the name(s) of the deployment artifact input parameter(s) through which the SMO can provide the names of the network attachment definitions (NADs) the SMO has created as base for the network interface the ExtCpd represents. The param name(s) and provided NAD name value(s) will be passed to the deployment tool when deploying the deployment artifacts. It is expected that the NADs themselves have been created prior to the deployment of the deployment artifacts. See notes 1, 2, and 3.
nadNamespace	no	string		Specifies, for an ExtCpd representing a secondary network interface, the name of the deployment artifact input parameter through which the SMO can provide the namespace where the NADs are located. The param name and provided namespace value will be passed to the deployment tool when deploying the deployment artifacts. Attribute may be omitted if the namespace is the same as the application namespace. See note 2.

NOTE 1: When the ExtCpd represent a network redundant/mated pair of sriov interfaces, there are references to 2 or 3 related NADs needed to be passed, while for other interface types only one NAD reference is needed to be passed.

NOTE 2: The format of the string content is specific for each different orchestration templating technology used (Helm, Teraform, etc.). Currently only a format for use with Helm charts is specified: "<helmchartname>:[<subchartname>.] 0 ... N[<parentparamname>.] 0 ... N<paramname>". Whether the optional parts of the format are present depends on how the parameter is declared in the helm chart. An example is: "chartName:subChart1.subChart2.subChart3.Parent1.Parent2.Parent3.parameter".

NOTE 3: A direct attached (passthrough) network interface, such as a sriov interface, attaches to a network via only one of the two switch planes in the infrastructure.

When using a direct attached network interface one therefore commonly in a pod uses a mated pair of sriov network attachments, where each interface attaches to the same network but via a different switchplane.

The application uses the mated pair of network interfaces as a single logical "switch-path-redundant" network interface – and this is represented by a single ExtCpd.

Also, there is a case where a third "bond" attachment interface is used in the pod, bonding the two direct interfaces so that the application does not need to handle the redundancy issues – application just uses the bond interface.

In this case all three attachments are together making up a logical "switch-path-redundant" network interface represented by a single ExtCpd.

When three NADs are used in the ExtCpd the NAD implementing the bond attachment interface is provided through the parameter indicated in the third place in the nadName property.

7.3.4.3 Definition

The syntax of the `tosca.datatypes.asd.ExtCpdParamMappings` data type is defined in annex A.

7.3.5 RequiredCustomResource

7.3.5.1 Description

Type name: **`tosca.datatypes.asd.RequiredCustomResource`**

This data type indicates the required custom resource types in the target environment, as defined in clause 6.7.

7.3.5.2 Properties

The properties of the `tosca.datatypes.asd.RequiredCustomResource` data type are specified in table 7.3.5.2-1.

Table 7.3.5.2-1: Properties

Name	Required	Type	Constraints	Description
kind	yes	string		The kind of the custom resource.
apiVersion	no	string		The api version of the custom resource. If not indicated, any api version of the custom resource is valid.

7.3.5.3 Definition

The syntax of the `tosca.datatypes.asd.RequiredCustomResource` data type is defined in annex A.

7.3.6 RequiredPlugin

7.3.6.1 Description

Type name: **`tosca.datatypes.asd.RequiredPlugin`**

This data type indicates the required plugins in the target environment, as defined in clause 6.7.

7.3.6.2 Properties

The properties of the `tosca.datatypes.asd.RequiredPlugin` data type are specified in table 7.3.6.2-1.

Table 7.3.6.2-1: Properties

Name	Required	Type	Constraints	Description
requiredPluginName	yes	string		The name of the required K8s plugin (e.g., multus).
requiredPluginVersion	no	string		The version of the required plugin (e.g., 3.8). If not indicated, any version of the plugin is valid.

7.3.6.3 Definition

The syntax of the `tosca.datatypes.asd.RequiredPlugin` data type is defined in annex A.

7.4 Artifact types

7.4.1 DeploymentItem

7.4.1.1 Description

Type name: **tosca.artifacts.asd.DeploymentItem**

This artifact contains the cloud native deployment file, as defined in clause 6.3.

7.4.1.2 Properties

The properties of the **tosca.artifact.asd.DeploymentItem** artifact type are specified in table 7.4.1.2-1.

Table 7.4.1.2-1: Properties

Name	Required	Type	Constraints	Description
deploymentItemId	yes	string		The identifier of this deployment item
artifactType	yes	string	Valid values: See YAML definition constraints	Specifies the artifact type.
deploymentOrder	no	integer		Specifies the deployment order of the deployment item. A lower value specifies that the deployment item belongs to an earlier deployment stage, i.e., needs to be installed prior to a deployment item with higher deploymentOrder values. If not specified, the deployment of the deployment itemt can be done in arbitrary order and decided by the SMO.
deploymentParameter	no	List of string		The list of parameters that can be overridden at deployment time (e.g., the list of parameters in the values.yaml which can be overridden at deployment time)

It can be noted that there is no artifact property corresponding to the artifactId attribute defined in clause 6.3. The reference to the file is conveyed with the mandatory ‘file’ keyname, which is supported in all TOSCA attribute definitions.

The artifact definition of the vendor specific deploymentItem shall contain the mandatory keynames ‘type’ and ‘file’ and the optional keyname ‘properties’ with the properties specified above. It may also contain the optional keyname ‘description’. All other optional artifact definition keynames listed in clause 3.6.7.1 in [4] shall not be used.

7.4.1.3 Definition

The syntax of the **tosca.artifacts.asd.DeploymentItem** is defined in annex A.

7.4.2 ExecutableImageItem

7.4.2.1 Description

Type name: **tosca.artifacts.asd.ExecutableImageItem**

This artifact contains the executable image file, as defined in clause 6.8.

7.4.2.2 Properties

The properties of the `tosca.artifact.asd.ExecutableImageItem` artifact type are specified in table 7.4.2.2-1.

Table 7.4.2.2-1: Properties

Name	Required	Type	Constraints	Description
<code>swImageName</code>	yes	string		Name of the software image
<code>swImageVersion</code>	yes	string		Version of the software image

It can be noted that there is no artifact property corresponding to the `swImageLocation` attribute defined in clause 6.8. The reference to the file is conveyed with the ‘file’ keyname, which is supported in all TOSCA attribute definitions.

The artifact definition of the vendor specific `executableImageItem` shall contain the mandatory keynames ‘type’ and ‘file’ and the optional keyname ‘properties’ with the properties specified above. It may also contain the optional keyname ‘description’. All other optional artifact definition keynames listed in clause 3.6.7.1 in [4] shall not be used.

7.4.2.3 Definition

The syntax of the `tosca.artifacts.asd.ExecutableImageItem` is defined in annex A.

7.5 Node types

7.5.1 Asd

7.5.1.1 Description

Type name: **`tosca.nodes.asd.Asd`**

The Asd node type represents the Application Service Descriptor information element, as defined in clause 6.2.

7.5.1.2 Properties

The properties of the `tosca.nodes.asd.Asd` node type are specified in table in table 7.5.1.2-1.

Table 7.5.1.2-1: Properties

Name	Required	Type	Constraints	Description
asdId	yes	string		Globally unique identifier of the ASD. See note 1.
asdVersion	yes	string		Identifies the version of the ASD. See note 2
asdProvider	yes	string		Provider of the application and of the ASD.
asdApplicationName	yes	string		Name to identify the application. Invariant for the application lifetime.
asdApplicationSwVersion	yes	string		Specifies the version of the application software. See note 3.
asdApplicationInfoName	no	string		Human readable name for the application. Can change during the ASD lifetime.
asdInfoDescription	no	string		Human readable description of the application. Can change during the ASD lifetime.
asdExtCpd	no	list of tosca.datatypes.asd.Ext CpdData		Describes the externally exposed connection points of the application.
enhancedClusterCapabilities	no	tosca.datatypes.asd.En hancedClusterCapabili ties		A list of expected capabilities of the target deployment cluster to aid placement of the application on a suitable cluster.
asdInvariantId	yes	string		Identifier of this descriptor in a version independent manner. This attribute is invariant across versions of ASD. See note 1.

NOTE 1: The value shall comply with an UUID format as specified in IETF RFC 4122 [3].

NOTE 2: Version changes when any change is made in the ASD.

NOTE 3: Version changes when the application software (software images or deploymentItem artifacts) changes

It can be noted that there is no property corresponding to the asdSchemaVersion attribute defined in clause 6.2. The version of the schema, i.e. the version of the standard specification that the ASD is compliant with, is declared in the import statement in the ASD service template.

A node template of type `tosca.nodes.asd.Asd` shall contain one or more artifacts of type `tosca.artifacts.asd.DeploymentItem` and may contain one or more artifacts of type `tosca.artifacts.asd.ExecutableImageItem`

The node template definition of the vendor specific asd node shall contain the mandatory keyname ‘type’ and the optional keynames ‘properties’ and ‘artifacts’. It may also contain the optional keynames ‘description’ and ‘metadata’. All other optional keynames defined in clause 3.8.3 in [4] shall not be used.

7.5.1.3 Definition

The syntax of the `tosca.nodes.asd.Asd` is defined in annex A.

7.6 ASD TOSCA service template

The ASD of a specific application is written in a YAML file which is called TOSCA service template in TOSCA terminology.

The service template shall contain the following elements in this order:

1. A ‘`tosca_definitions_version`’ key-value pair indicating the applicable version of the TOSCA specification, which is `tosca_simple_yaml_1_3` in this version of the ASD specification.

```
tosca_definitions_version: toscasimpleyaml_1_3
```

2. Optionally a description of the content of the service template as a key-value pair, i.e. the key ‘description’ followed by colon and a string of text:

```
description: This ASD is ...
```

3. An import statement referencing the TOSCA type definitions file containing the standard O-RAN ASD type definitions specified in Annex A.

```
imports:
  - oran_asd_types_v02.00.00
```

The file containing the standard O-RAN ASD type definitions shall not be included in the package delivered by the solution provider. If it is included it shall be ignored. This is in order to avoid intended or unintended modifications in the standard types. The file with the standard type definitions is assumed to be well known and its location accessible to the entity processing the ASD service template.

4. A topology template

In turn, the topology template shall consist of:

- 4.1 The key ‘topology_template’ followed by colon
- 4.2 Optionally a description as a key-value pair , i.e. the key ‘description’ followed by colon and a string of text
- 4.3 The key ‘node_templates’ followed by colon
Both the ‘description’ key, if present’, and the ‘node_templates’ key shall be indented with respect to the ‘topology_template’ key. It is recommended to use two-space indentation.
- 4.4 A node template definition of type `tosca.nodes.asd.Asd`
The node template definition shall be indented with respect to the ‘node_templates’ key.

The node template definition shall include at least one artifact definition of type `tosca.artifacts.asd.DeploymentItem` and may contain one or more artifact definitions of type `tosca.artifacts.asd.ExecutableImageItem`.

7.7 Example

```
tosca_definitions_version: toska_simple_yaml_1_3

imports:
  - oran_asd_types_v02.00.00 # O-RAN standard ASD types

topology_template:
  node_templates:
    LoggingNFDeployment: # name of node template chosen by the solution provider
      type: toska.nodes.asd.Asd
      properties:
        asdId: b1bb0ce7-ebca-4fa7-95ed-4840d70a1177
        asdInvariantId: c1bb0ab8-deab-4fa7-95ed-4840d70a3574
        asdVersion: 1.0.0
        asdProvider: MyCompanyName
        asdApplicationName: MultiPurposeLogger
        asdApplicationSwVersion: 1.0.0
        asdInfoDescription: Provides logging functionality to O-CU NF.
        asdExtCpd:
          - id: 01
```

```

description: This extCpd exposes the management service of the NF Deployment
virtualLinkRequirement: OAM
networkInterfaceRealizationRequirements:
  trunkmode: false
  ipam: infraProvided
  interfaceType: kernel.netdev
  interfaceRedundancy: infraProvided
inputParamMappings:
  nadName: LoggerIHC:ExtCp1.Nad # LoggerIHC is the name of the Helm chart chosen by the
solution provider
enhancedClusterCapabilities:
  minKernelVersion: 4.15.0
  requiredCustomResource:
    - kind: Redis
      apiVersion: kubedb.com/v1alpha1
    - kind: NetworkAttachmentDefinition
artifacts:
  LoggerHelmChart: # name of artifact chosen by solution provider
    type: tosca.artifacts.asd.DeploymentItem
    file: Artifacts/Helm/LoggerIHC
    properties:
      deploymentItemId: 01
      artifactType: helm_chart
      deploymentParameter:
        - logfilename
  LoggerSwImage: # name of artifact chosen by solution provider
    type: tosca.artifacts.asd.ExecutableImageItem
    file: https://MyCompany.com/images/Logger\_01\_01\_00
    properties:
      swImageName: Logger
      swImageVersion: 01.01.00

```

Annex A (normative): YAML types definition

```

tosca_definitions_version: tosca_simple_yaml_1_3
description: ORAN ASD types definitions version 02.00.00
metadata:
  template_name: oran_asd_types_v02.00.00
  template_author: ORAN
  template_version: 02.00.00
data_types:
  tosca.datatypes.asd.ExtCpdData:
    derived_from: tosca.datatypes.Root
    description: >
      describes the externally exposed connection points of the application.
    properties:
      id:
        type: string
        description: The identifier of this ExtCpdData.
        required: true
      description:
        type: string

```

```

    description: >
        Describes the service exposed by the external connection point
        represented by this ExtCpdData.
    required: true
    virtualLinkRequirement:
        type: string
        description: >
            Refers in an abstract way to the network that the external
            connection point shall be exposed for (for example, OAM, EndUser, backhaul,
            LI, etc).
        required: true
    networkInterfaceRealizationRequirements:
        type: tosc.datatypes.asd.NetworkInterfaceRequirements
        description: >
            Details container implementation specific requirements on the
            NetworkAttachmentDefinition.
        required: false
    inputParamMappings:
        type: tosc.datatypes.asd.ExtCpdParamMappings
        description: >
            Information on what parameters are required to be provided to the
            deployment tools for the external connection point.
        required: false
    resourceMapping:
        type: string
        description: >
            Resource name for the cloud native resource manifest, as specified
            in the cloud native descriptor.
        required: false

tosc.datatypes.asd.EnhancedClusterCapabilities:
    derived_from: tosc.datatypes.Root
    description: >
        describes expected capabilities of the target deployment cluster.
    properties:
        minKernelVersion:
            type: string
            description: >
                Describes the minimal required Kernel version, e.g., 4.15.0. Coded
                as displayed by linux command uname -r.
            required: true
        requiredKernelModule:
            type: list
            description: >
                Required kernel modules are coded as listed by linux lsmod command,
                e.g., ip6_tables, cryptd, nf_nat etc.
        entry_schema:
            type: string
            required: false
        conflictingKernelModule:
            type: list
            description: >

```

Kernel modules that shall not be present in the target environment. The kernel modules are coded as listed by linux lsmod command, e.g., ip6_tables, cryptd, nf_nat etc.

```
entry_schema:
  type: string
  required: false
requiredCustomResource:
  type: list
  description: >
```

List of the required custom resource types in the target environment. The list shall include those custom resource types which are not delivered with the application.

```
entry_schema:
  type: tosc.datatypes.asd.RequiredCustomResource
  required: false
clusterLabel:
  type: list
  description: >
```

This attribute indicates the required O-Cloud Node Cluster capabilities.

```
entry_schema:
  type: string
  required: false
```

```
requiredPlugin:
  type: list
  description: >
```

A list of the names and versions of the required K8s plugin.

```
entry_schema:
  type: tosc.datatypes.asd.RequiredPlugin
  required: false
```

```
tosca.datatypes.asd.NetworkInterfaceRequirements:
```

```
  derived_from: tosc.datatypes.Root
  description: >
```

details related to secondary networks that attach the OS containers to the logical or physical networks.

```
  properties:
```

```
    trunkMode:
      type: boolean
      description: >
```

Specifies whether the interface is capable of carrying traffic for multiple VLANs.

```
      required: true
      default: false
```

```
    ipam:
      type: string
      description: >
```

Specifies which mode is used for the IP address assignment.

```
      required: true
      constraints:
```

```
        - valid_values: [ infraProvided, orchestrated, userManaged ]
```

```
      default: infraProvided
```

```

interfaceType:
  type: string
  description: >
    Specifies the type of network interface. Different types of
interfaces are supported by different CNIs.
  required: true
  constraints:
    - valid_values: [ kernel.netdev, direct.userspace,
direct.kerneldriver, direct.bond, userspace ]
  default: kernel.netdev
interfaceOption:
  type: list
  description: >
    vNIC configurations the network interface is verified to work with.
This attribute is applicable for interfaces of type userspace.
  required: false
  entry_schema:
    type: string
    constraints:
      - valid_values: [ virtio, memif ]
interfaceRedundancy:
  type: string
  description: >
    Method required from the infrastructure to provide redundancy for
the interface.
  required: false
  constraints:
    - valid_values: [ infraProvided, left, right, activeActiveBond,
activePassiveBond, activePassiveL3 ]

tosca.datatypes.asd.ExtCpdParamMappings:
  derived_from: toska.datatypes.Root
  description: >
    details required information on what parameters to be provided to the
deployment tools for the ExtCpd instance.
  properties:
    loadbalancerIP:
      type: string
      description: >
        Specifies the name of the deployment artifact input parameter
through which the SMO can configure the loadbalancerIP parameter of the K8s
service or ingress controller that the ExtCpd represents.
      required: false
    externalIPs:
      type: string
      description: >
        Specifies the name of the deployment artifact input parameter
through which the SMO can configure the externalIPs parameter of the K8s
service or ingress controller, or the pod network interface annotation, that
the ExtCpd represents.
      required: false
  nadName:

```

```

    type: list
    description: >
        Specifies, for an ExtCpd representing a secondary network
interface, the name(s) of the deployment artifact input parameter(s) through
which the SMO can provide the names of the network attachment definitions
(NADs) the SMO has created as base for the network interface the ExtCpd
represents. It is expected that the NADs themselves have been created prior
to the deployment of the deployment artifacts.
    required: false
    entry_schema:
        type: string
    nadNamespace:
        type: string
    description: >
        Specifies, for an ExtCpd representing a secondary network
interface, the name of the deployment artifact input parameter through which
the SMO can provide the namespace where the NADs are located. Attribute may
be omitted if the namespace is same as the application namespace.
    required: false

tosca.datatypes.asd.RequiredCustomResource:
    derived_from: tosca.datatypes.Root
    description: >
        Indicates the required custom resource types in the target
environment.
    properties:
        kind:
            type: string
            description: The kind of the custom resource.
            required: true
        apiVersion:
            type: string
            description: >
                The api version of the custom resource. If not indicated, any api
version of the custom resource is valid.
            required: false

tosca.datatypes.asd.RequiredPlugin:
    derived_from: tosca.datatypes.Root
    description: >
        Indicates the required plugins in the target environment.
    properties:
        requiredPluginName:
            type: string
            description: The name of the required K8s plugin.
            required: true
        requiredPluginVersion:
            type: string
            description: >
                The version of the required plugin. If not indicated, any version
of the plugin is valid.

```



```
required: false
```

```
artifact_types:
```

```
  tosca.artifacts.asd.DeploymentItem:
```

```
    derived_from: tosca.artifacts.Root
```

```
    description: >
```

```
      Artifact containing the cloud native deployment file.
```

```
    properties:
```

```
      deploymentItemId:
```

```
        type: string
```

```
        description: The identifier of this deployment item.
```

```
        required: true
```

```
      artifactType:
```

```
        type: string
```

```
        description: Specifies the artifact type.
```

```
        required: true
```

```
        constraints:
```

```
          - valid_values: [ helm_chart, helmfile, crd, terraform ]
```

```
      deploymentOrder:
```

```
        type: integer
```

```
        description: >
```

Specifies the deployment order of the deployment item. A lower value specifies that the deployment item belongs to an earlier deployment stage, i.e., needs to be installed prior to a deployment item with higher deploymentOrder values. If not specified, the deployment of the deployment item can be done in arbitrary order and decided by the SMO.

```
        required: false
```

```
        constraints:
```

```
          - greater_or_equal: 0
```

```
      deploymentParameter:
```

```
        type: list
```

```
        description: >
```

The list of parameters that can be overridden at deployment time (e.g., the list of parameters in the values.yaml which can be overridden at deployment time)

```
        required: false
```

```
        entry_schema:
```

```
          type: string
```

```
  tosca.artifacts.asd.ExecutableImageItem:
```

```
    derived_from: tosca.artifacts.Root
```

```
    description: >
```

```
      Artifact containing the executable image file.
```

```
    properties:
```

```
      swImageName:
```

```
        type: string
```

```
        description: Name of the software image.
```

```
        required: true
```

```
      swImageVersion:
```

```
        type: string
```

```

        description: Version of the software image.
        required: true

node_types:

  toska.nodes.asd.Asd:
    derived_from: toska.nodes.Root
    description: >
      Node type to be used to define the vendor specific node template that
      represent the vendor ASD.
    properties:
      asdId:
        type: string # UUID
        description: Globally unique identifier of this ASD.
        required: true
      asdVersion:
        type: string
        description: Identifies the version of the ASD.
        required: true
      asdProvider:
        type: string
        description: Provider of the application and of the ASD.
        required: true
      asdApplicationName:
        type: string
        description: >
          Name to identify the application. Invariant for the application
          lifetime.
        required: true
      asdApplicationSwVersion:
        type: string
        description: >
          Specifies the version of the application software.
        required: true
      asdApplicationInfoName:
        type: string
        description: >
          Human readable name for the application. Can change during the ASD
          lifetime.
        required: false
      asdInfoDescription:
        type: string
        description: >
          Human readable description for the application. Can change during
          the ASD lifetime.
        required: false
      asdExtCpd:
        type: list
        description: >
          Describes the externally exposed connection points of the
          application.
        entry_schema:
          type: toska.datatypes.asd.ExtCpdData
          required: false
      enhancedClusterCapabilities:

```

```
type: tosa.datatypes.asd.EnhancedClusterCapabilities
description: >
  A list of expected capabilities of the target deployment cluster to
aid placement of the application on a suitable cluster.
  required: false
asdInvariantId:
  type: string # UUID
  description: >
    Identifier of this descriptor in a version independent manner. This
attribute is invariant across versions of ASD.
  required: true
```

Annex (informative): Change History

Date	Revision	Description
2023.04.17	00.00.01	Initial TS skeleton
	00.00.02	Editor's notes with references to clauses in O-RAN.WG6.AppLCM-Deployment-R003-v01.00 for content, and informative document reference added.
2023.08.23	00.00.03	Implemented CRs: NOK-2023.06.07-WG6-CR-ASD_conventions (NOK-0054 rev 1) NOK-2023.06.07-WG6-CR-ASD_overview-02 (NOK-0055 rev 2)
2023.09.07	00.00.04	Implemented CRs: NOK-2023.06.07-WG6-CR-ASD_application_service_descriptor_clauses-v05 (NOK-0056 rev 5) ERI-2023.06.14-WG6-CR-0015-NetworkInterfaceRealizationRequirements-v03 (ERI-0015 rev 3)
2023.10.27	00.00.05	Implemented CRs: ERI-2023.09.26-WG6-CR-0019-ASD-corrections-v04 ERI-2023.10.04-WG6-CR-0020-ASD-empty-clauses-and-cleanup-v02
2023.11.07	00.00.06	Copyright changed to 2024
2023.11.15	00.00.07	Addresses comments from Thinh at November 2023 WG Approval Comment Wiki - Cloudification and Orchestration Workgroup - Confluence (atlassian.net) Implements three changes from ERI-2023.09.26-WG6-CR-0019-ASD-corrections-v04 that were missed when implementing the CR in v.00.00.05.
2023.11.21	00.00.08	Addresses comments from Joan at November 2023 WG Approval Comment Wiki - Cloudification and Orchestration Workgroup - Confluence (atlassian.net)
2023.11.21	00.00.09	Final resolution of Joan's comments according to agreement in WG6 meeting on 2023.11.21. November 2023 WG Approval Comment Wiki - Cloudification and Orchestration Workgroup - Confluence (atlassian.net)
	01.00	Final version 01.00
2024.10.16	01.00.01	Implements CRs: ERI-2024.09.05-WG6-CR-0041-ASD-Stage3-Skeleton-v02 ERI-2024.09.16-WG6-CR-0044-ASD-Stage3-Datatypes-v02
2024.11.13	01.00.02	Implemented CRs: ERI-2024.10.02-WG6-CR-0045-ASD-Stage2-SwImageDescriptor-v05 ERI-2024.10.07-WG6-CR-0046-ASD-Stage2-AsdVersions-v03 ERI-2024.10.07-WG6-CR-0047-ASD-Stage3-Nodetypes-v01 ERI-2024.10.07-WG6-CR-0048-ASD-Stage3-ArtifactTypes-v02 ERI-2024.10.07-WG6-CR-0049-ASD-UseOfToscaClauses-v02 ERI-2024.10.07-WG6-CR-0050-ASD-EditorialCorrections-v01
2025.02.04	01.00.03	Implemented CRs: ERI-2024.11.20-WG6-CR-0051-ASD-ServiceTemplate-v01 ERI-2024.11.28-WG6-CR-0052-ASD-ServiceTemplateExample-v02 ERI-2024.12.11-WG6-CR-0055-ASD-ResourceMapping-v01
2025.03.04	01.00.04	Implemented CRs: ERI-2025.01.23-WG6-CR-0065-ASD-AdditionsAndCorrections-v05
2025.03.27	02.00	Final version 02.00